

- (a) Write the formulas similar to Equations 4 for the center of mass $(\bar{x}, \bar{y}, \bar{z})$ of a thin wire in the shape of a space curve C if the wire has density function $\rho(x, y, z)$.
- (b) Find the center of mass of a wire in the shape of the helix $x = 2 \sin t$, $y = 2 \cos t$, $z = 3t$, $0 \leq t \leq 2\pi$, if the density is a constant k .

Solution

Calculate the x -coordinate of the center of mass. *sec 2 x-coordinate*

$$\begin{aligned}
 \bar{x} &= \frac{\int x \, dm}{\int dm} = \frac{\int_C x(\rho \, ds)}{\int_C \rho \, ds} = \frac{\int_0^{2\pi} x(t)k \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt}{\int_0^{2\pi} k \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt} \\
 &= \frac{k \int_0^{2\pi} (2 \sin t) \sqrt{(2 \cos t)^2 + (-2 \sin t)^2 + (3)^2} dt}{k \int_0^{2\pi} \sqrt{(2 \cos t)^2 + (-2 \sin t)^2 + (3)^2} dt} \\
 &= \frac{\int_0^{2\pi} (2 \sin t) \sqrt{13} \, dt}{\int_0^{2\pi} \sqrt{13} \, dt} \\
 &= \frac{2 \int_0^{2\pi} \sin t \, dt}{\int_0^{2\pi} dt} \\
 &= 0
 \end{aligned}$$

Calculate the z -coordinate of the center of mass.

-Coordinate

C_1

$$\begin{aligned}\bar{z} &= \frac{\int z \, dm}{\int dm} = \frac{\int_C z(\rho \, ds)}{\int_C \rho \, ds} = \frac{\int_0^{2\pi} z(t)k \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt}{\int_0^{2\pi} k \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt} \\&= \frac{k \int_0^{2\pi} (3t) \sqrt{(2 \cos t)^2 + (-2 \sin t)^2 + (3)^2} dt}{k \int_0^{2\pi} \sqrt{(2 \cos t)^2 + (-2 \sin t)^2 + (3)^2} dt} \\&= \frac{\int_0^{2\pi} (3t) \sqrt{13} dt}{\int_0^{2\pi} \sqrt{13} dt} \\&= \frac{3 \int_0^{2\pi} t dt}{\int_0^{2\pi} dt} \\&= 3\pi\end{aligned}$$

Therefore, the center of mass is $(0, 0, 3\pi)$.

5. A thin wire is bent into the shape of a semicircle $x^2 + y^2 = 4, y \geq 0$. If the linear density is $k|x|y$, for some constant $k > 0$, find the mass and center of mass of the wire.

Solution: We may parametrize the wire $\mathcal{W}$ via $x = 2 \cos t, y = 2 \sin t, 0 \leq t \leq \pi$.

$$\text{The mass is } \int_{\mathcal{W}} k|x|y \, ds = \int_0^\pi k(2|\cos t|)(2 \sin t) \sqrt{(\pm \sin t)^2 + (\cos t)^2} \, dt =$$

$$4k \int_0^\pi \sin t |\cos t| \, dt = 8k \int_0^{\pi/2} \sin t \cos t \, dt = 4k \sin^2 t \Big|_0^{\pi/2} = 4k(1^2 - 0^2) = 4k.$$

$$M_y = \int_{\mathcal{W}} x \cdot k|x|y \, ds = \int_0^\pi k(2|\cos t|)(2|\cos t|)(2 \sin t) \sqrt{(\pm \sin t)^2 + (\cos t)^2} \, dt =$$

$$8k \int_0^\pi \sin t \cos^2 t \, dt = \frac{8k \cos^3 t}{3} \Big|_0^\pi = 0.$$

$$M_x = \int_{\mathcal{W}} y \cdot k|x|y \, ds = \int_0^\pi k(2 \sin t)(2|\cos t|)(2 \sin t) \sqrt{(\pm \sin t)^2 + (\cos t)^2} \, dt =$$

$$8k \int_0^\pi \sin^2 t |\cos t| \, dt = 16k \int_0^{\pi/2} \sin^2 t \cos t \, dt = \frac{16k \sin^3 t}{3} \Big|_0^{\pi/2} = \frac{16k}{3} - 0 = \frac{16k}{3}.$$

$$\text{Hence, } \bar{x} = \frac{M_y}{M} = \frac{0}{4k} = 0, \bar{y} = \frac{M_x}{M} = \frac{16k/3}{4k} = \frac{4}{3} \text{ and the center of mass is at } (0, \frac{4}{3}).$$

A spring is made of a thin wire twisted into the shape of a circular helix $x=2 \cos t$, $y=2 \sin t$, $z=t$. Find the mass of two turns of the spring if the wire has constant mass density.

$$\rho \int_0^{4\pi} \sqrt{4(\sin^2 t + \cos^2 t) + 1} dt$$

$$\sqrt{5} \rho \int_0^{4\pi} dt$$

$$\sqrt{5} \rho [t]_0^{4\pi}$$

$$\boxed{4\sqrt{5} \pi \rho}$$

Find the mass and center of mass of a wire in the shape of the helix $x = t$, $y = \cos t$, $z = \sin t$, $0 \leq t \leq 2\pi$, if the density at any point is equal to the square of the distance from the origin.

Solution

The linear density of the wire is

$$\rho(x, y, z) = (\sqrt{x^2 + y^2 + z^2})^2 = x^2 + y^2 + z^2.$$

Integrate this density over the length of the wire to obtain the total mass.

$$\begin{aligned} M &= \int_C \rho \, ds = \int_0^{2\pi} \{[x(t)]^2 + [y(t)]^2 + [z(t)]^2\} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \, dt \\ &= \int_0^{2\pi} (t^2 + \cos^2 t + \sin^2 t) \sqrt{(1)^2 + (-\sin t)^2 + (\cos t)^2} \, dt \\ &= \int_0^{2\pi} (t^2 + 1) \sqrt{2} \, dt \\ &= \sqrt{2} \left(\frac{t^3}{3} + t \right) \Big|_0^{2\pi} \\ &= \sqrt{2} \left(\frac{8\pi^3}{3} + 2\pi \right) \\ &= \frac{2\sqrt{2}\pi}{3} (4\pi^2 + 3) \end{aligned}$$

Calculate the x -coordinate of the center of mass.

$$\begin{aligned} \bar{x} &= \frac{\int x \, dm}{\int dm} = \frac{\int_C x(\rho \, ds)}{M} \\ &= \frac{3}{2\sqrt{2}\pi(4\pi^2 + 3)} \int_0^{2\pi} x(t) \{[x(t)]^2 + [y(t)]^2 + [z(t)]^2\} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \, dt \\ &= \frac{3}{2\sqrt{2}\pi(4\pi^2 + 3)} \int_0^{2\pi} t(t^2 + \cos^2 t + \sin^2 t) \sqrt{(1)^2 + (-\sin t)^2 + (\cos t)^2} \, dt \\ &= \frac{3}{2\sqrt{2}\pi(4\pi^2 + 3)} \int_0^{2\pi} t(t^2 + 1) \sqrt{2} \, dt \\ &= \frac{3}{2\pi(4\pi^2 + 3)} (2\pi^2 + 4\pi^4) \\ &= \frac{3\pi(1 + 2\pi^2)}{4\pi^2 + 3} \approx 4.60 \end{aligned}$$